

Week 4 · Laboratory Solutions

- Reference answers, not the only correct ones. The checker tests the physics, not the diagram.
- Read them after attempting the problem.

Building and checking

```
cd lectures/W04_simulink/solutions
W04_S1_los          % builds W04_S1.slx – both laws in one model
W04_S_expected      % regenerates the expected-result figures

W04_check(1,'W04_S1')
W04_check(2,'W04_S1')
W04_check(3,'W04_S1')
```

All three pass and `check_overlaps('W04_S1')` is **0**.

```
W04 problem 1
pi_p on leg 1          0.0000 (expected  0.0000 +- 1e-09 rad) PASS
settled cross-track error -0.0306 (expected  0.0000 +- 0.05 m) PASS
settled heading command  0.0038 (expected  0.0000 +- 0.01 rad) PASS
it started 18 m off the line 18.0000 (expected 18.0000 +- 0.05 m) PASS

W04 problem 2
LOS   worst |y_e| after 90 s    0.0945 m
atan2 worst |y_e| after 90 s    3.7812 m
atan2 still reaches the waypoint 1.4216 (expected  0.0000 +- 6 m) PASS

W04 problem 3
measured crab angle beta_c    22.4615 deg
settled offset y_e           3.3011 (expected  3.3074 +- 0.15 m) PASS
the prediction Delta tan(beta_c) 3.3074 (expected  3.3011 +- 0.15 m) PASS
heading error while offset persists -0.0036 (expected  0.0000 +- 0.5 deg) PASS
```

Problem 3 is the one worth pausing on. The prediction $\Delta \tan \beta_c = 3.3074$ m and the measurement **3.3011** m agree to **six millimetres**, and they do so while the heading error is -0.004° . Section 4-7's claim is not an approximation.

The three decisions worth defending

	The choice	Why the alternative is worse
One rotation, both errors	x_e^p and y_e^p from the same matrix product	A distance formula plus a sign test gives y_e^p and throws x_e^p away. Section 4-6's switching test needs the along-track coordinate, and computing it separately invites the two to disagree
<code>atan2</code> for π_p , <code>atan</code> for the correction	two different functions on purpose	A leg can point into any quadrant, so π_p needs both arguments. The aim-point vector cannot leave the right half-plane of $\{p\}$ because $\Delta > 0$, so the correction does not
The command is not wrapped	wrapping happens in the autopilot, on the error	A wrapped command is harmless here and wrong in general: differentiate it for a feed-forward term and the 2π jump becomes an impulse

Why both laws live in one block

The comparison in Problem 2 is only worth making if the plant, the autopilot and the allocation are **bit-for-bit identical** between the two runs. A flag inside one guidance block guarantees that; two models do not.

It is the same reason the lecture's own `W04_guidance.slx` runs four laws through one autopilot bank, and the same reason Week 2 held three controllers in one model.

The measurement window, and why it matters

Problem 2's first draft compared the two laws from $t = 60$ s and the LOS vessel **failed its own check** at **0.44** m. Nothing was wrong with the model: at **60** s the LOS vessel is still finishing its approach, so the comparison was between a transient and a steady state.

Moving the window to $t \geq 90$ s — the same window Problem 1 uses — gives **0.09** m against **3.78** m.

The general lesson is the one this course applies to its own figures: **a measured number means nothing without the sentence that says what was measured**. Choosing a window that flatters one side is as much an error as computing the wrong quantity.

What is deliberately missing

No waypoint switching, no ILOS, no ALOS. Leg 1 only.

Switching is section 4-6 and needs the along-track error the guidance already computes; the two adaptive laws are sections 4-8 and 4-9. Problem 3 exists to make the reader want them: it ends with a vessel that is holding its commanded heading perfectly and is still **3.3** m from where it was asked to be.